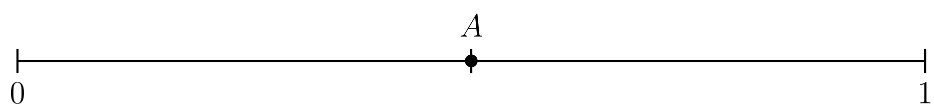
Placing Fractions on a Number Line

Naming the fraction at a marked point, placing a fraction on a partitioned line, and comparing two fractions on the line

How to read a fraction on a number line.

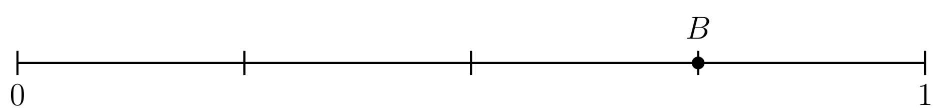
- Count the *equal parts* between 0 and 1 — not the tick marks. That count is the bottom number (the denominator).
- Count how many of those parts you travel from 0 to reach the point. That count is the top number (the numerator).
- Always start counting at 0, and always count whole parts.

1. The line below goes from 0 to 1. It is cut into 2 equal parts. Point A sits on the first tick mark after 0. Write the fraction that names point A.



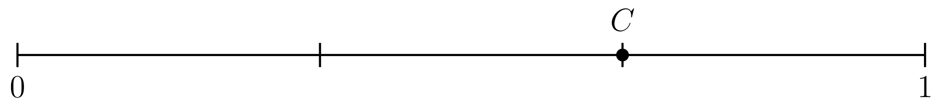
Answer: ____________________

2. This line goes from 0 to 1 and is cut into 4 equal parts. Count the equal parts from 0 up to point B, then write the fraction that names point B.



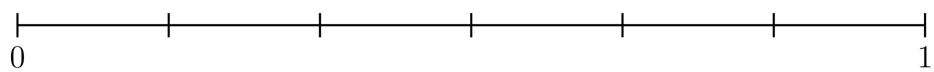
Answer: ____________________

3. This line goes from 0 to 1 and is cut into 3 equal parts. Write the fraction that names point C.



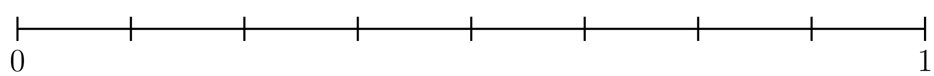
Answer: ____________________

4. This line goes from 0 to 1 and is cut into 6 equal parts. Put a dot on $\frac{4}{6}$ and label it P. Then write the *simpler* fraction that names the very same point.



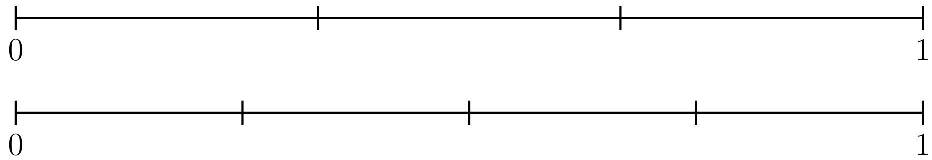
Answer: ____________________

5. This line goes from 0 to 1 and is cut into 8 equal parts. Put a dot on $\frac{8}{8}$. Which whole number names that same point?



Answer: ____________________

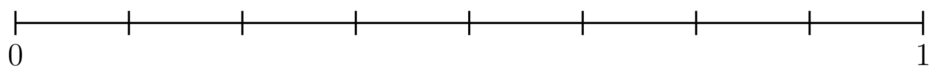
6. The top line is cut into thirds and the bottom line is cut into fourths. Mark $\frac{2}{3}$ on the top line and $\frac{2}{4}$ on the bottom line. Which one lands farther from 0? Write $>$ or $<$ in the blank so the sentence is true.



$\frac{2}{3}$ ______ $\frac{2}{4}$

Answer: ________________

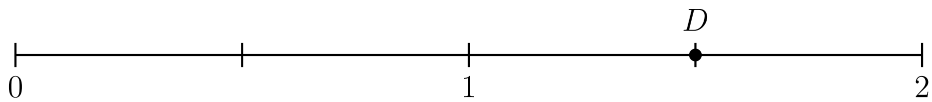
7. This line is cut into 8 equal parts. Mark $\frac{5}{8}$ on it. Then mark $\frac{3}{4}$ on the same line — remember that one fourth is the same length as two eighths. Write $>$ or $<$ in the blank so the sentence is true.



$\frac{5}{8}$ ______ $\frac{3}{4}$

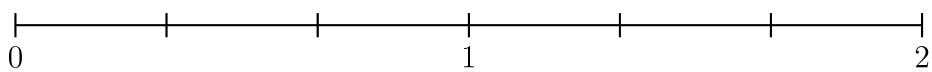
Answer: ________________

8. This line does not stop at 1: it runs all the way to 2, and every whole is cut into halves. Write the fraction that names point D.



Answer: ________________

9. This line runs from 0 to 2 and every whole is cut into thirds. Put a dot on $\frac{5}{3}$ and label it E. How far past 1 does E sit? Write that distance as a fraction of one whole.



Answer: ________________

10. Ellie says that $\frac{1}{3}$ is farther from 0 than $\frac{1}{2}$, because 3 is a bigger number than 2. The top line is cut into thirds and the bottom line into halves. Mark $\frac{1}{3}$ on the top line and $\frac{1}{2}$ on the bottom line, then write one sentence explaining what Ellie is getting wrong.